

ARHAM MAHMOOD

Artificial Intelligence Engineer | Computer Vision | NLP & Generative AI

Rawalpindi, Pakistan | +92 3151547906 | mah.arhamper@gmail.com

GitHub: ArhamMahmood | LinkedIn: arham-mahmood-175b832a1

Summary

Aspiring AI Engineer and BS Artificial Intelligence candidate at NUTECH with hands-on experience in Computer Vision, Natural Language Processing, and Generative AI. Demonstrated ability to develop and fine tune deep learning models through a national research collaboration with NCAI, where I contributed to a real-world AI Retail Security & Theft Detection system. Proficient in Python, PyTorch, TensorFlow, and LangChain, with a consistent track record of building end-to-end AI solutions. Actively seeking an internship or entry-level AI/ML role where I can apply my skills, learn from industry professionals, and contribute meaningfully to impactful projects.

Skills

Programming Languages: Python, SQL, Java, C++

AI / ML Frameworks: PyTorch, TensorFlow, Scikit-learn, LangChain, Hugging Face Transformers

Computer Vision & NLP: OpenCV, NLTK, SpaCy, BERT, GPT-2

Data & Visualization: Pandas, NumPy, Matplotlib, Seaborn

Tools & Platforms: Git, GitHub, Gradio

Work history

AI Research Intern

Jul 2025 – Aug 2025

National Center of Artificial Intelligence (NCAI), Islamabad, Pakistan

- Collaborated with NCAI researchers as part of a 2-month internship tied to the FYP project AI-Retail Security & Theft Detection
- Received expert mentorship on model selection, architecture decisions, and best practices for real-world Computer Vision pipelines
- Designed and trained two baseline deep learning models from scratch one for shoplifting detection and one for firearm/gun detection using annotated video datasets
- Delivered functional base models that were subsequently used as the foundation for further training, fine tuning, and integration into the final FYP system

Projects

AI-Retail Security & Theft Detection (FYP)

2025 – Present

YOLOv8 | SlowFast | PyTorch | OpenCV | Computer Vision | Team of 3

- Developed an end-to-end surveillance system using a hybrid YOLOv8 + SlowFast architecture for spatial temporal action recognition.
- Curated a unique dataset of real-world videos from local Pakistani marts to solve the lack of localized shoplifting data.
- Engineered a modular pipeline designed to ingest video streams and automatically dispatch real-time alerts to management using Telegram Bot.

Text Autocompletion

NLP | BERT | GPT-2 | Gradio

- Developed a Gradio-based web application combining BERT for next-word prediction and GPT-2 for full-sentence generation from user input
- Demonstrates practical integration of transformer-based models for real-time language generation tasks

Gender Classification from Speech

Audio Processing | Feature Extraction | Scikit-learn

- Built a binary classification model that identifies speaker gender from audio signals by extracting vocal features including pitch, formants, and MFCCs
- Applied signal processing techniques and trained a supervised classifier, demonstrating cross-domain AI skills beyond text and vision

Education

Bachelor of Science in Artificial Intelligence

Nov 2022 – Present

National University of Technology (NUTECH), Islamabad, Pakistan

- Relevant Coursework: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, Data Structures & Algorithms
- Final Year Project: AI-Retail Security & Theft Detection a real-time computer vision system for shoplifting and weapon detection, developed in collaboration with NCAI

FSc Pre-Engineering

2020 – 2022

Army Public School (APS), Ordnance Road

Certifications

- Fundamentals of AI Agents Using RAG and LangChain
- Neural Networks and Deep Learning (DeepLearning.AI)
- Improving Deep Neural Networks: Hyperparameter Tuning & Optimization
- Generative AI: Prompt Engineering Basics